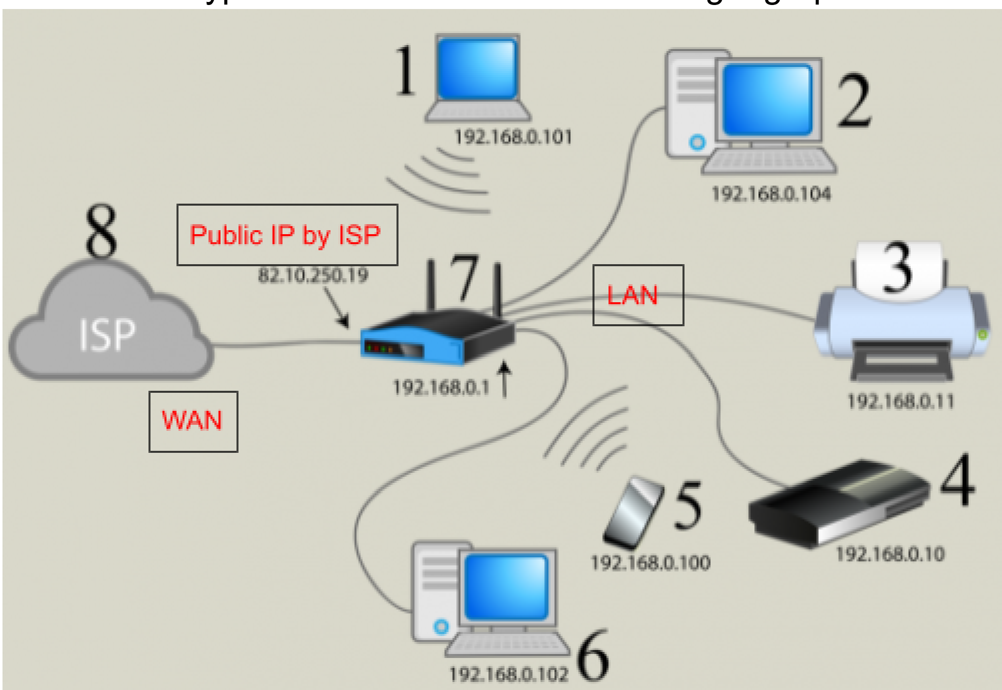
NETWORKING

Title: <u>Network fundamentals Concepts</u>

For a devops engineer knowing about networking is crucial. I started to go through basic concepts as well as required skills for devops. In this document I will cover fundamental concepts of networking.

Network: Connection between at least two network devices/nodes. Network grows as a number of devices are connected like computers, routers, switches etc. Networks are categorized into different types on the basis of its size and geographical area.



LAN: If communication between systems routes(spans) within local or restricted areas this network will be called local area network. Here are a few main characteristics of LAN.

- Devices are directly connected instead of connecting through the internet.
- Mostly connected via cable (twisted pair or ethernet) and wifi
- For example two systems from one system to another system in different cities connected through a direct cable is an example of LAN.
- The faster and the most secure network.

WAN / INTERNET: If devices communicate via the internet irrespective of their locations. Like if two systems communicate via facebook or gmail within a room this will be WAN. Here are few characteristics of WAN

- This network is establish with fiber-optics
- Slower than LAN and MAN
- More vulnerable network
- Connected through routers.

MAN: This is a kind of network like a company that has multiple offices at different places of a city or different cities, but they are directly connected to each other so this will be MAN. Its characteristics lie between LAN and WAN.

IP Addresses

- Unique identifier of network devices
- Can be of one type (IPv4 or IPv6)
- IPv4 IP address is based on 32 bit(4-octets). Each octet range from 0-255 (2^8)
- IP can be **public**(accessible from internet) and **private** (cannot accessible from internet)

IPs are categorized into different classes as mentioned below.

Class A

- **Range**: `1.0.0.0` to `126.255.255.255`
- **Subnet Mask**: `255.0.0.0` (`/8`)
- **Purpose**: Large networks (e.g., governments, corporations).

Class B

- **Range**: `128.0.0.0` to `191.255.255.255`
- **Subnet Mask**: `255.255.0.0` (`/16`)
- **Purpose**: Medium-sized networks (e.g., universities).

Class C

- **Range**: `192.0.0.0` to `223.255.255.255`
- **Subnet Mask**: `255.255.255.0` (`/24`)
- **Purpose**: Small networks (e.g., home/office LANs).

Class D

- **Range**: `224.0.0.0` to `239.255.255.255`
- **Purpose**: Multicasting (not for devices).

Class E

- **Range**: `240.0.0.0` to `255.255.255.255`
- **Purpose**: Reserved (experimental).

Key Notes:

- **127.0.0.0/8** is reserved for loopback (e.g., `127.0.0.1` = localhost).
- Modern networks use **CIDR** (e.g., `192.168.1.0/24`) instead of classes for flexible subnetting.

Please correct me if I mentioned something that needs to be corrected.

---------------------- Thank you 🙂 ----------------------